

Problem Set 2



D. Jason Koskinen
koskinen@nbi.ku.dk

Advanced Methods in Applied Statistics
Feb - Mar 2026

Info

- The submission is:
 - A write-up as a PDF document, which includes any plots, diagrams, tables, pictures, and explanations
 - In a separate “file”, submit all code used to derive the results
 - Tarball, zipped directory, lots of individual files w/ self-explanatory titles, etc.
 - Do NOT include lines of code in your write-up. If results are dependent on coding choices then include those comments in the write-up.
- Include any original data files or how the data was accessed
 - If you use a internet scraping tool, note the date when you retrieved the data
 - If you can save the data to a file, do so and submit the data file. There is no need to change the format, e.g. HTML, XML, txt, JSON...

Problem 1 (3.5 points)

- Running is a hobby for those wishing to simultaneously have a bad time and be dismayed at their fitness level. Even so, the following dataset revolves around a 10k race in which people *willingly* spent time and money for the 'privilege' of participating.
- Because these masochists are trying to run as fast as possible, it seems reasonable to use a Generalized Extreme Value (GEV) distribution as an underlying hypothesis for the probability distribution function of the finishing times.
- The data file can be found at <https://www.nbi.dk/~koskinen/Teaching/data/Liverpool10k2026.csv> with the time format of HH:MM:SS

Generalized Extreme Value

- The PDF of the Generalized Extreme Value function is:

$$f(x | \mu, \sigma, \xi) = \frac{1}{\sigma} \left[1 + \xi \left(\frac{x - \mu}{\sigma} \right) \right]^{-\frac{1}{\xi} - 1} \exp \left\{ - \left[1 + \xi \left(\frac{x - \mu}{\sigma} \right) \right]^{-\frac{1}{\xi}} \right\}$$

- There are 3 parameters (μ, σ, ξ) that we will fit to the 10k data
- The above form for the GEV is specifically for cases where $\xi \neq 0$. Be mindful that there are other constraints for the parameters (μ, σ, ξ) such that certain combinations will produce an ill-defined output which might be problematic during any minimization/maximization process.

Problem 1a (1.5 pts.)

- Using the GEV function, what are the best-fit values of (μ, σ, ξ) after fitting the function to the 10k data?
 - Use an unbinned likelihood approach (or whatever you feel is appropriate) to get the best-fit values $(\hat{\mu}, \hat{\sigma}, \hat{\xi})$
 - Include all entries of the 10k time data set, regardless of gender, name, or running club affiliation
- What is the -LLH (negative ln-likelihood) of the data using your best-fit values?
 - Technically, you can get the best-fit values $(\hat{\mu}, \hat{\sigma}, \hat{\xi})$ by any approach you want, but here you must provide the -LLH value using those best-fit values. So if you used a minimization approach based on likelihood minimization, this value should be very easy to produce.
- Plot a histogram of the 10k data, **and** overlay a 'scaled' GEV function using the best-fit values so that the histogram and function can be easily compared visually.
 - x-axis should be 10k times, and y-axis should be frequency
 - The scaled GEV PDF should be a smooth line, whereas the histogram should be a histogram
 - Do **NOT** normalize the histogram to make it small enough to match the fitted GEV PDF. Instead scale the GEV PDF up so that the shape and location is relatively easy to compare to the histogram.

Problem 1b (1 pt.)

- Using the GEV function with your best-fit parameters from Problem 1a, what is the probability from the GEV PDF of running a time slower than that of "Tanyusha, SomethingAboutGoats"?
- From the actual data, what is the fraction of runners that ran slower than "Tanyusha, SomethingAboutGoats"?
- Provide a brief discussion about whether these two values (the GEV Probability vs. the data-based fraction) are in reasonable agreement with each other or if they are in possible disagreement.

Problem 1c (1 pt.)

- Let us now focus on the fast times, i.e. the tail of the distribution going towards lower times.
- Using values of (μ, σ, ξ) that are $(2980, 607.5, -0.26)$ plot a scaled version of the Cumulative Distribution Function of the General Extreme Value function, along with the histogram of the 10k data.
 - Extremely similar to 1a, but now instead of the PDF we are plotting a scaled version of the CDF.
- Using the given values of $(2980, 607.5, -0.26)$, what is the probability from the GEV distribution of getting a time faster than the fastest time in the 10k data set?

Problem 2 (2.5 points)

- Bayesian statistics

- The likelihood will be the PDF for a gamma function $\Gamma(\alpha, \beta)$, which is defined as:

$$f(x|\alpha, \beta) = \frac{\beta^\alpha x^{\alpha-1} e^{-\beta x}}{\Gamma(\alpha)}$$

- For this problem $\alpha = 2$ and $\beta = 2$
- The prior will be a step function defined as:

$$f(x) = \begin{cases} 1/N, & 0 < x \leq 1.6 \\ 2.66/N, & 1.6 < x \leq 2.1 \\ 0, & x \leq 0 \text{ or } x > 2.1 \end{cases}$$

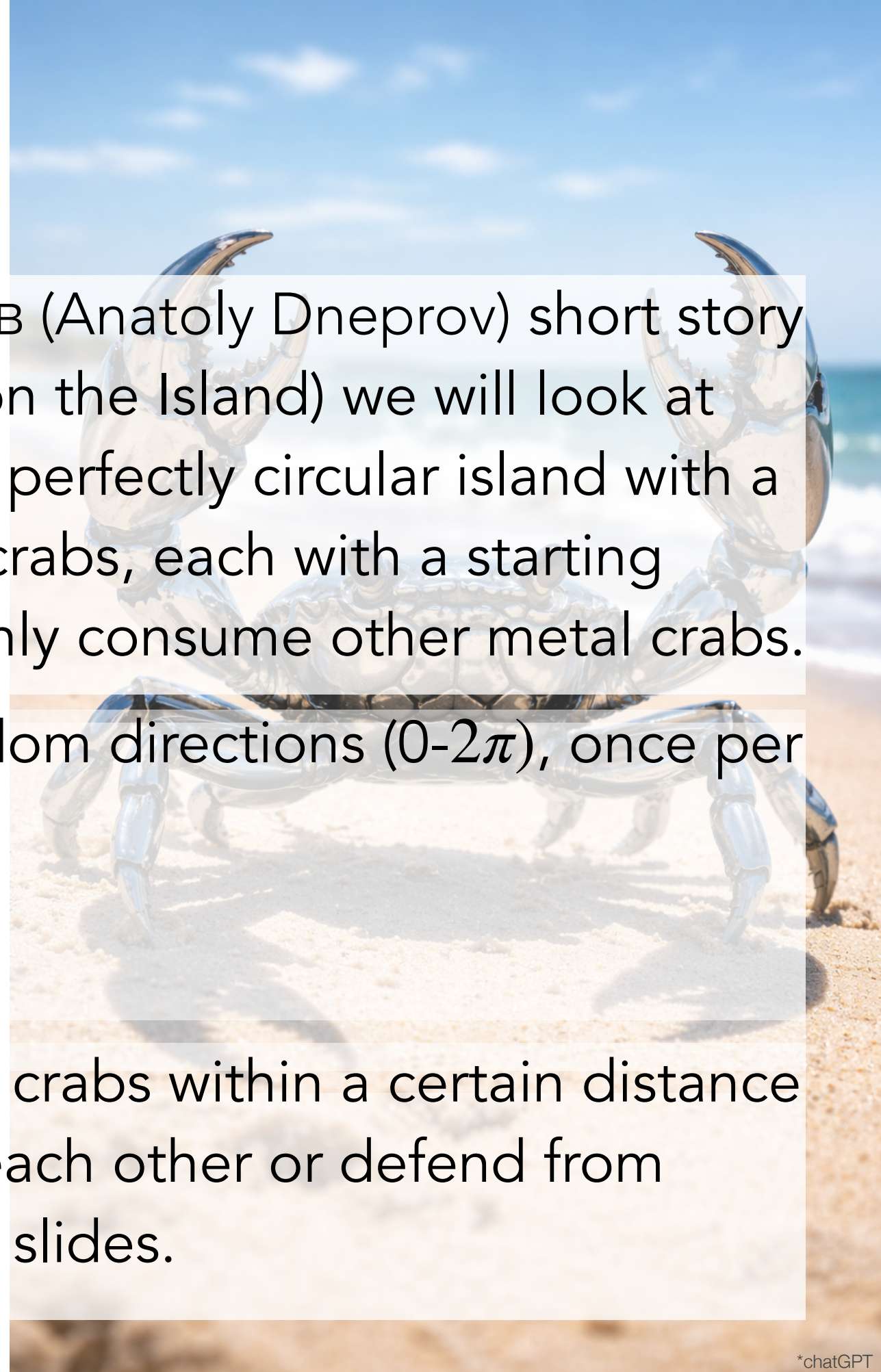
for a value of N which normalizes $f(x)$

Problem 2 continued

- Plot the likelihood, prior, and bayesian posterior on the same plot over the range of $0 < x < 4$
 - All 3 functions should be properly normalized
 - All 3 plots should use a different color scheme than the Matplotlib default. I.e. no usage of blue, orange, or green even if you aren't using matplotlib for plotting
 - All axes should have correct labels and the plot should include an appropriate legend
- Using the bayesian posterior, what is the most likely value of x , i.e. $\hat{x}$?
- If $g(x)$ is the bayesian posterior, what is the value of $g(\hat{x})$?

Problem 3 (4 points)

- Inspired by the Анатолій Дніпров (Anatoly Dneprov) short story Крабы идут по острову (Crabs on the Island) we will look at population evolution. Imagine a perfectly circular island with a 5 km radius inhabited by metal crabs, each with a starting mass of 1 kg. The metal crabs only consume other metal crabs.
- All crabs move in individual random directions ($0-2\pi$), once per day until:
 - they move 200 meters, or
 - they reach the edge of the island
- At the end of the day, any metal crabs within a certain distance of another crab will try and eat each other or defend from being eaten. Details are on later slides.

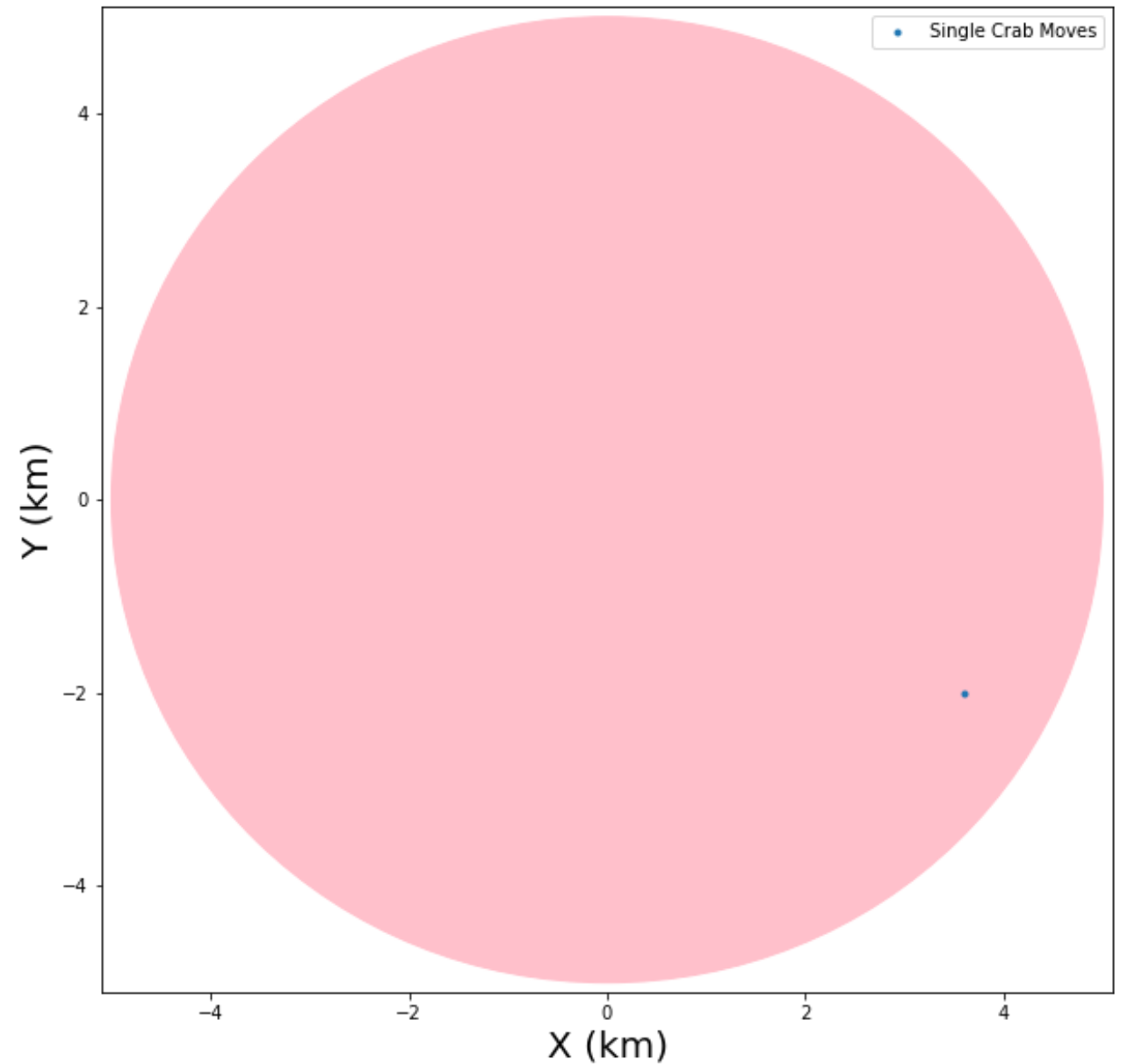


Crab Movement

- Consider the crabs to move instantaneously (for the simplicity of this Monte Carlo simulation).
- If a crab reaches the island edge they will continue to move the next day, i.e. they do not stop forever upon reaching the island's edge.
- A crab will move up to 200 m *if possible*. For example, if it is 50.2 m away from the island boundary and chooses a direction directly towards the island boundary, the crab will move 50.2 m that day and then stop.
- The center of the circular island is at (0 km, 0 km).

Problem 3a (0.5 points)

- For simplicity, we consider only a single crab on the island for this part of the problem. No other cannibal metal crabs... yet.
- Show a plot of a single crab starting at (3.6 km, -2.0 km) and moving for 200 days.
 - Put a single point for each stopping position of the crab
 - Example figure ONLY includes the starting point, i.e. day=0
 - There should be a total of 200 points, or 201 if you include the starting position.



Problem 3b (1 point)

- For simplicity, we continue considering only a single crab on the island, and simulate the total distance travelled before reaching the edge of the island.
- Run 501 pseudo-experiments with a single crab starting at the location (3.6 km, -2.0 km) and make a histogram of the 501 distances the crab travels before arriving at the edge of the island.
 - The histogram x-axis range axis range should go from 0-200 km, and the bin widths should be 2 km.
 - This is the total travel distance of the crab, and NOT the straight line distance between the starting point and the edge of the island where the crab ultimately stops.

Problem 3 Crab Battle Details

- Crabs that end the day within 175 m of each other will battle. The larger crab will consume the smaller crab with odds of $\frac{M_{larger}^2}{1kg} : \frac{M_{smaller}^2}{1kg}$.
 - For example, a 5 kg crab eating a 2 kg crab has 25:4 odds, i.e. $25/(25+4)=86.2\%$. The smaller crab has a $4/(25+4)=13.8\%$ of surviving, but might get eaten the next day (life is difficult for small crabs).
 - If the smaller crab defends, then both crabs continue to exist.
 - Crabs of equal mass have 1:1 odds of one crab consuming the other. It is arbitrary which one is considered 'larger'.
- If the larger crab eats the smaller crab its mass increases equal to the smaller crab it consumed. The position on the island of the 'winning' crab remains the same as where it stopped for the day, despite the battle. In the case of the smaller crab surviving, then both crabs stay in the same position as where they stopped for the day.
- Although rare, multiple crabs can end their day within 175 m of multiple other crabs. In such scenarios, the battles occur in order of shortest distance to largest distance.

Problem 3 Crab Battle Tips

- Make sure that the sum of metal crab mass on the island is always 20 kg. If it's ever not equal to 20 kg, then there's a bug in your code.
- Eaten crab cannot battle. If a crab is consumed, somehow keep track that it cannot battle anymore. There are many, many, many ways to do the 'book keeping' for this. For example,
 - Change the mass of the consumed crab to 0 and the position to -infinity.
 - Remove the eaten crab from the array or list after the battle in which they are eaten.
 - Include a data element in the crab array, list, or class which notes whether it has 1) battled already that day, and/or 2) whether it has been eaten.

Problem 3c (1.5 points)

- There are 20 metal crabs on the island with the starting positions provided in the file at <https://www.nbi.dk/~koskinen/Teaching/data/CrabStartPositions.txt>.
 - The first column is the x position and the second column is the y position.
 - Each row is the starting position of an individual crab.
- After 200 days:
 - What is the mostly likely number of individual crabs that remain alive?
 - What is the most likely mass of the largest crab?

Problem 3d (1 points)

- We will continue using the same 20 crab on the island with the starting positions provided in the file at <https://www.nbi.dk/~koskinen/Teaching/data/CrabStartPositions.txt>
- Show the distribution of days it takes until only 10 crabs remain alive
- What is the 1σ (i.e. 68.27%) confidence interval on the number of days at which only 10 crabs remain?
 - If the distribution is asymmetric, then calculate the lower 1σ bound as the value where the lower tail contains $(100\%-68.27\%)/2$ of the distribution. Same thing for the upper 1σ bound calculated using the higher tail.
 - N.B. review the lecture *Parameter Estimation and Uncertainty* for a reminder about confidence intervals.